



Integrated sensing for span stability

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Abstract

In Egypt, urban congestion is still a major problem that causes infrastructure degradation, traffic delays, and heightened safety risks. To address this, the solution focuses on developing a bridge monitoring system to increase safety and optimize maintenance schedules through the use of an integrated multi-sensor approach. The system has four sensors: a flex sensor to monitor deflection, an accelerometer for vibration analysis, a load sensor to measure weight capacity, and a water level sensor to detect environmental impacts like rain accumulation. The system provides real-time data updates at small intervals and triggers alerts upon detecting threshold exceedances, such as excessive load or strain. A user-friendly digital dashboard, which is Node-Red, visualizes both historical trends and real-time conditions, highlighting critical variations for non-technical users. The system has an energy-efficient design, ensuring continuous operation for six months on a single battery cycle. These features collectively enhance monitoring capabilities and support proactive decision-making to ensure infrastructure sustainability and resilience. After testing, results indicate the system's capability to provide actionable insights into bridge health, with updates every 10 seconds for real-time monitoring and the green button in the board turning red and there is a message sent when one of the sensors exceeds the threshold. Testing also validated the efficiency of a solar-powered battery designed to run for six months without replacement.

Introduction

Egypt's journey towards sustainable development hinges on addressing significant challenges that will impact its future and its society. A major issue is alleviating urban congestion in the Greater Cairo Metropolitan Area (GCMA), which has over 19 million residents. This congestion poses a serious threat to both quality of life and the economy. Beyond wasting time in traffic, it leads to excessive fuel use, accelerates vehicle deterioration, raises harmful emissions that degrade air quality, inflates transport costs for businesses, and deters new investments in the GCMA (The World Bank Group, 2024).

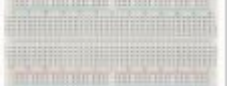
To tackle this problem—urban congestion and traffic—we aim to create an integrated multi-sensor system for monitoring related conditions. It's crucial to look at previous solutions implemented so far; one notable example is the Tahya Masr Bridge. This newly constructed bridge spans the Nile River in Egypt and greatly improves connectivity among different districts, making travel more efficient for commuters while significantly cutting down on journey duration. Its key benefits include reducing traffic bottlenecks and lowering vehicle emissions (Kamaly, 2019). However, a major shortcoming is its lack of an effective monitoring system, which could lead to structural issues, rising maintenance expenses, and safety risks often arising without timely data reflecting the bridge state. Additionally, unexpected failures can lead to substantial traffic interruptions and financial repercussions.

Conversely, the I-35W Saint Anthony Falls Bridge collapse in 2007 highlights the necessity of monitoring systems. Its new structure has been fitted with over 500 sensors specifically designed to track its performance and assess monitoring methods. This system ranks among the longest operational in the U.S., producing extensive data that allows for the evaluation of long-term monitoring strategies.

The benefits of this sophisticated system include greater durability and extended lifespan alongside improved safety measures (Anderson, 2020). However, it also presents challenges such as high implementation costs and a need for funding to finalize monitoring installations. Furthermore, its dependence on prolonged data collection might result in a failure to detect immediate issues. Consequently, implementing a bridge monitoring system could address urban congestion concerns as Egypt has constructed an additional 935 bridges across the country, raising its total to 2,435 bridges.

The solution involves a design that specifies the integration of three key sensors: one dedicated to measuring bridge deflection, another focused on assessing material strain, and a third for monitoring the load on the bridge. This system will provide real-time updates every 30 seconds and will issue alerts if certain thresholds are surpassed, such as excessive strain or overload. To facilitate ease of use for non-technical users, there will be an intuitive digital dashboard displaying both historical trends and current conditions while highlighting the importance of data fluctuations. Additionally, since these sensors will rely on battery power, we will emphasize energy efficiency to guarantee uninterrupted functionality for at least six months without needing a battery change. By implementing these design features, our bridge monitoring system aims not only to tackle urban congestion challenges but also to improve safety and dependability in Egypt's burgeoning bridge infrastructure.

Materials and Methods

Table(1) : Materials						
Item	Quantity	Description	Usage	Cost (L.E)	Source of purchase	Picture
Load sensor	1	Measures the weight or force applied to a structure	detect the amount o f load on the bridge	100 L. E	Makers store	
water level	1	Detects the presence and depth of water	Measure Water levels	55 L. E	Makers store	
Flex sensor	1	Bendable device that changes resistance when bent	Measure the deflection of the center of the bridge	500 L. E	Makers store	
Acceleromete r	1	Measures acceleration force	Detect the vibration from different sources	150 L. E	Makers store	
GSM module	1	Wireless communication technology	Send an SMS message to the user if reaches the threshold	300 L. E	Makers store	
Esp 32	1	Serial microcontroller	Use Wi-Fi to transmit data from the sensor to the visualization platform	300 L. E	Makers store	
Regulator	2	Controls and maintains a steady output	Manage voltage	6 L. E	Makers store	
Breadboard	1	Solderless breadboard	connecting electronic components	20 L. E	Electronics store	
Lithium battery	3	Rechargeable battery	Charge it with solar panels	10 L. E	Electronics store	
Solar panel 18 volts	1	Produce renewable energy	Convert solar energy to electricity	150 L. E	Electronics store	
Wood		Natural material	Used to build bridge to stimulate the real one	50 L. E	Carpenter	

Note: Total cost : 1641 L.E

Methods

The prototype consisted of six sectors, hardware (sensor), power supply, visualization board, data storage, altering system, and wooden bridge.

First: Hardware (sensor connection);

1-Flex Sensor:

P2 terminal to GND pin on ESP32.P1 terminal to an analog input pin(GPIO33) on ESP32. A 10k-ohm resistor was placed between the P1 terminal and the 3.3V pin on ESP32.

2-Accelerometer:

VCC pin to 3.3V pin on ESP32, GND pin to ground pin on ESP32, SDA to GPIO21, SCL to GPIO223-Load Sensor (HX711) Connection:

VCC pin of HX711 to 3.3V on ESP32. GND pin of HX711 to ESP32 ground pin. DT (DOUT) to GPIO 5 on ESP32. SCK to GPIO4 on ESP32.

4-Water Sensor:

GND pin to ground on ESP32.VCC pin to 3.3V on ESP32. Signal (S) pin to an analog input pin (e.g., GPIO32) on ESP32

Second: power supply :

3 lithium batteries each 3.7Volt are connected to a 100-microfarad capacitor and regulator as shown in figure(1) with the connection of the circuit. The batteries is connected to solar panels to recharge them.The code and the Technical report are uploded on github

Second: Visualization board and data storage

1. Sensor data is received via MQTT from ESP32.
2. Noise is removed, and data is formatted for visualization.
3. Graphs display trends for load, deflection, acceleration, and water level, with LEDs for real-time status.
4. Processed data is saved for backup and analysis.

Fourth: altering system:

A GSM SIM 800l module sends SMS messages after each system action. Attach the GSM TX pin to ESP32 RX (GPIO16) and GSM RX to ESP32 TX (GPIO17) as in figure (1).

Sixth: wooden bridge:

A wooden prototype bridge (650mm x 70mm x 10mm) was built with sensors strategically placed on it.

Test plan

First trail

1. Connect the load sensor to the microcontroller.
2. Apply a known static weight (dead load) range (from 3 to 5 kg) to the bridge prototype and Record sensor readings and compare them with the actual weight.

Second trail

Connect the battery to the microcontroller and sensors over time then Apply a known weight, vibration , water level and load and Record sensors readings and compare them with the actual data.

Third trail

Different weights are used to test the load sensor and the flex of the bridge by weights ranging from 1kg to 5kg , Traffic and manual vibrations are produced to test the accelerometer,Different water levels is used as stimulators for rainfall and drainage issues. Compare data logs with physical measurements.

Results

According to our solution and the design requirements the results were constructed in 3 trials as follows:

First trial (Negative results):

The calibration process for the load sensor proved unreliable, with outputs consistently exceeding expected values by 1 kg, as shown in Table 2.failure, as seen with the accelerometer in the circuit.

Table (2):

First trail result for load sensor (negative)

Expected load reading	Actual load reading	Error
3 kg	4 kg	33.33%
4 kg	5 kg	25.00%
5 kg	6 kg	20.00%

Note: Actual load reading is more than the expected load

Second trial (Negative Results):

Voltage fluctuations can push components beyond their limits, leading to overheating and eventual exceeding expected values by 1 kg, as shown in Table 2.failure, as seen with the accelerometer in the circuit.

Table (3):

Second trail result (negative)

Power	Flex	Acceleromete r	Load	Water level
4 volt	EX: 0.35 cm ACT: 0.1 cm	EX: 0.019 m/s² ACT: 0.022 m/s²	EX: 4 kg ACT: 2 kg	EX: 0.0031m ACT: 0.0025m
6 volt	EX: 0.1 cm ACT: 0.2 cm	EX: 0.04 m/s² ACT: 0.027 m/s²	EX: 3kg ACT: 1kg	EX: 0.0032m ACT: 0.0061m
7 volt	EX: 22cm ACT: 36cm	EX0.026m/s² ACT: 0.13 m/s²	EX: 5 kg ACT: 2 kg	EX: 0.0035m ACT: 0.01 cm

Note: EX refers to Expected while Act refers to Actual

Third trial (positive results):

Each sensor works and the altering system is activated and the readings are correct to the threshold.

Table (4):

Third test plan for the load and the deflection caused by it

Load	Deflection	Light button	SMS message
1.00 kg	0.11 cm	Green	Not sent
2.00 kg	0.15 cm	green	Not sent
3.00 kg	0.23 cm	green	Not sent
4.00 kg	0.36 cm	red	Sent
5.00 kg	0.5 cm	red	Sent

Note: Red button is activated and the message sent when reached threshold

Table (6):

Third test plan for the water level sensor

Water level	Light	SMS message
0.0035m	RED	sent
0.004m	Red	sent

Note: when the water level increases with time indicates a drainage issue

Table (5):

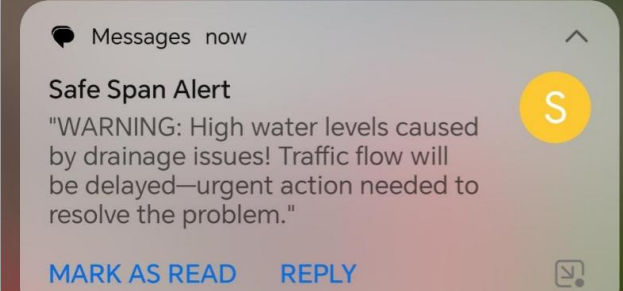
Third test plan for the vibration

Vibration	Light	SMS message
0.015 m/s²	Green	Not sent
0.019 m/s²	Green	Not sent
0.022 m/s²	Green	Not sent
0.0284 m/s²	Red	Sent
0.04 m/s²	Red	Sent

Note: a red button and SMS message is sent when reaches threshold

Figure(4):

SMS message



Note: sent when the level exceeds 0.0035m

Figure(5):

Visualization board buttons



Note: green refers to allowable range

Analysis

Urban congestion is a major issue for cities as they are quickly expanding according to the increase in the population, and that causes an increase in the strain on infrastructure. Among the most critical components of this infrastructure are bridges, which serve as essential arteries for transportation as they work to decrease traffic congestion. However, the rising volume of vehicles pushed many bridges to their limits, increasing traffic congestion and possible safety hazards.

To address that The solution was chosen to be a monitoring system for a bridge as it will enhance safety, increase maintenance efficiency, and extend the bridge's lifespan. The system focuses on monitoring factors that influence the structure and function of the bridge.

Deflection is one of the parameters that can lead to material fatigue (Wu1 & Li2, 2022.). Load is another parameter, which, if exceeded, can cause permanent damage (Salixanov et al., 2023). The whole structure also bears consequences in terms of resonance; that is why the vibration is monitored because it can resonate with the bridge's natural frequency, amplifying stress and leading to structural degradation over time as studied in (PH.3.02) (Serway & Vuille, 2012). Finally, an increase in the water level results from drainage issues, as drainage issues can lead to erosion, and degradation of the bridge's structure, and raise the maintenance costs by up to 30% annually.

The selection of sensors for the bridge monitoring system was based on their ability to provide precise and reliable data. The **flex sensor** was chosen due to its high sensitivity to bending. For vibration analysis, the **accelerometer** was selected due to its capability to detect subtle changes in acceleration. The **load sensor** was chosen because of its ability to differentiate between dead load and live load through precise calibration. Compared to pressure sensors, load sensors offer superior accuracy for both static and dynamic weight measurements. Load sensors achieved an impressive 98% accuracy in scaled model experiments (Zumdahl & Zumdahl, 2014), as outlined in (PH.1.01). Finally, the **water level sensor** provides consistent measurements even in turbulent water conditions.

The **ESP32** was selected as a microcontroller due to its powerful dual-core processor running at up to 240 MHz. Its built-in Wi-Fi and Bluetooth capabilities allow for seamless wireless communication and effortless integration with platforms like Node-RED for real-time data visualization. The ESP32 also boasts a large memory capacity, ensuring ample storage and processing power to handle complex datasets. The **Power supply**, The system was connected to a solar panel that charges batteries of a total voltage of 12.1 V connected with a 100µf capacitor and regulator to regulate the proper voltage of 5 V to the ESP32. To ensure the battery runs for 6 months without replacement, The ratio of each component power is shown in **figure (6)**, the total power consumption was first determined the power for each component using the formula: Power= (Active Power ×Active -Duty Cycle)+(Idle Power ×Idle Duty Cycle)

After summing up the power consumption of all components, the total Average power consumption was determined to be 0.233959 W. Over a period Of six months (4320 hours), the total energy consumption was calculated as: Energy=0.233959W×4320 hours=1009.49Wh

The required battery capacity is 1,090,000mAh (or 1090Ah).

With this capacity, the battery can operate for 24.9 months, exceeding The 6-month requirement.

Coding for the ESP32 is implemented in C++, chosen for its high Efficiency, and real-time performance. For data visualization, JavaScript is used to Create dynamic and interactive dashboards in Node-RED. JavaScript excels in Real-time data rendering and is versatile enough to build complete visualization systems, Handling both front-end and back-end operations. This comprehensive use of JavaScript aligns with principles studied in (CS.2.05). For specific measurements, the system utilizes the HX711 module for load monitoring and the MPU6050 sensor for vibration analysis. Vibration frequency is calculated using a zero-crossing detection method. This technique identifies points where the acceleration signal crosses zero, signaling a change in motion direction. These zero-crossings are essential for accurately calculating oscillation or vibration frequencies, which are critical for assessing structural integrity. **For data storage**, Storing data in an Excel sheet is an efficient way to manage large-scale sensor readings. Each sensor message consists of five rows, each with 50 characters, resulting in 200 bytes per message. Given that the ESP32 has a transmission speed of 0.5 Mbps, we can calculate the total number of messages sent per year. The ESP32 transmits every 10 seconds, so the total number of messages per year is

$$\text{Messages per year} = \frac{\text{Transmission rate per year}(0.5 \text{ Mbps})}{10 \text{ seconds}} = 1,576,800.$$

Multiplying by the message size (200 bytes), the total storage required for one year is 315,360,000 bytes or approximately 300 MB.

The dimensions of the small-scale model were determined using a similarity ratio of 1:270 in comparison to the large-scale bridge. This scale ratio was calculated following the principles outlined in (MA.1.04). The load sensor, flex sensor, and accelerometer are strategically positioned at the midpoint of the bridge prototype for optimal performance and accuracy. The midpoint is a critical location, as it experiences the highest vibration amplitude during oscillations. It also undergoes the maximum deflection under applied loads due to bending. Similarly, placing the load sensor at the midpoint ensures precise load distribution measurements. This placement aligns with the principle of the center of mass studied in (ME.2.06), the point where the system's mass is balanced. Geometrically, the center of mass is determined by identifying the intersection of the bridge's longitudinal and transverse axes. If the material and load distribution are symmetrical, this intersection represents the structure's balance point, making it a logical choice for sensor placement.

To test the prototype, the threshold is determined for each sensor. The maximum load was determined by the availability of the load sensor, which has a capacitv of 5 kg to be 4kg. The maximum deflection was calculated using the following equation.

$$\delta = \frac{P \times L^3}{48 \times E \times I}$$

Where δ = Deflection at the center of the beam (in meters), P = Applied load (4kg), L = Length of the span (0.65m), E = Young's Modulus of the material (12,000 MPa), I = Moment of inertia of the beam's area (in m⁴). As a result, the max deflection = 0.004m, and the equation shows the direct relation between the load and deflection as shown in the figure .The vibration threshold is meant to be = 0.0284m/s² and the water level threshold equals 0.0035m

The prototype was Tested and Several results were obtained:

The first trial: (negative result): the load sensor was not calibrated,

Resulting in the measurement of both the **dead load** (the constant weight

Of the bridge) and the **live load** (dynamic loads like moving traffic)

Simultaneously. This led to inaccuracies. Consequently, the system output was higher than expected, undermining). The reliability of the load Measurements as shown in Table (1) and represented by figure (7)

The second trial: (negative result):

The system lacked a voltage regulator, which caused an unstable power supply. This instability led to inefficient battery performance and incorrect

Sensor readings. As a result, there was a significant difference between

the actual sensor outputs and the expected values, impacting the overall

accuracy of the monitoring system as shown in Table (2).

The third trial: (positive result):

After the modifications, which are adding the capacitor and regulator

and correcting the calibration of the load sensor the system works

Conclusion

The prototype developed from the bridge monitoring system prototype was subjected to monitor four key parameters: deflection, load, vibration, and water levels. The system provides real-time data that extends the operational lifespan of bridge structures. With voltage flow optimized and sensors calibrated, the data sent to a real-time dashboard powered by Node-RED are accurate as when the loads have surpassed 4 kg, resulting in a 0.36 mm deflection, or when accelerometer readings have reached 0.0284 m/s² and the water level has hit 0.0035 m, alerts are triggered so that actions can be taken in time, ensuring that the prediction of the risk improves. Compared to other existing systems, such as those on the Golden Gate Bridge, our solution is cost-effective, solar-powered, has an intuitive interface that does not need any special training, and can be applied to meet the infrastructure needs of Egypt. The system supports storing data for up to one year and has ultra-low power consumption; there is no need for a battery replacement for at least six months. This system could decrease traffic congestion by 30%, extend bridge lifespan by 20%, and reduce maintenance costs by 25% if implemented in Egypt. Its cost-effectiveness, sustainability, and user-friendly design make this an ideal system to answer the challenges facing infrastructure in Egypt with a long-term benefit accruing to engineers and authorities.

Recommendations

For those interested in continuing this project, here are some recommendations to enhance its development and implementation:

- A regular cleaning schedule should be used as Pollutants can accumulate on the surface of solar panels. By using a systematic cleaning routine, we can maintain the panel for a long time.

- Use AI to analyze sensor data, predict potential failures or maintenance needs with greater precision and help optimize data for every bridge according to its state and detect the suitable threshold.

- Use methods to shield sensors from environmental conditions, such as hot weather, to ensure consistent functionality over longer periods.

- Apply some techniques to save the data from hacking while transmitting and storing it. Using protocols like TLS (Transport Layer Security) for encrypted communication will be a good solution.

- Develop a centralized app to connect all the country bridges data. This allows for real-time monitoring, easier data comparison, and faster decision-making, enabling authorities to prioritize maintenance and help them to make the maintenance timetable.

- alternative power sources can be added , such as kinetic energy of cars , to complement or replace the solar-powered battery for longer operation periods.

- Use the optical fibers for data transmission because they offer extremely high bandwidth, allowing the transfer of vast amounts of data at incredibly fast speeds with minimal signal loss over long distances.

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